Simulation Retirement & Transition Module (SRTM)

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Independent Methodological Authority

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Canonical Definition

Simulation Retirement & Transition Module (SRTM) is the fifth and final Core Module of the Simulation Lifecycle & Resimulation Standard (SLRS) within SIMULOS® — Simulation Governance Architecture.

SRTM governs the controlled retirement, supersession, replacement, transition, archival disposition, and residual-use status of simulations that can no longer legitimately continue under their existing lifecycle state.

The Module establishes when a simulation should cease active use, when its governed role should transfer to a successor simulation, how dependencies and simulation-derived evidence must be treated during transition, and what historical simulation assets may remain available for explicitly bounded purposes.

SRTM prevents obsolete, superseded, materially drifted, unsupported, or retired simulations from continuing to influence decisions merely because they remain technically accessible.

Its task is:

to ensure that the end of active simulation use becomes a governed lifecycle state rather than an informal disappearance, silent replacement, or uncontrolled continuation.

Architectural Position

SRTM is the fifth and final module within SLRS.

It follows:

SRSM — Resimulation Scope Module.

The complete SLRS module sequence is:

SLMM — Lifecycle Monitoring → SDRM — Drift Recognition → SRGM — Resimulation Trigger → SRSM — Resimulation Scope → SRTM — Retirement & Transition

SRTM closes the active lifecycle where continued resimulation is no longer the appropriate response.

It may produce three principal lifecycle directions:

Continue → existing governed simulation remains active.

Transition / Replace → simulation responsibility moves toward a successor simulation basis.

Retire → simulation ceases active governed use.

Internal Flow Position

Lifecycle Monitoring Record Recognized Simulation Drift Resimulation Trigger Resimulation Scope Determination → Continuation Feasibility Assessment → Retirement / Transition Condition Assessment → Dependency Identification → Successor Simulation Identification where applicable → Transition Boundary Definition → Residual Use Determination → Evidence Consequence Identification → Lifecycle State Assignment → Retirement / Transition Record → Archived / Superseded / Retired Simulation State

Where a successor simulation is established:

Retired or Superseded Simulation → Successor Simulation Basis → Applicable SIMULOS® Lifecycle Re-Entry Operational Space

SRTM governs the lifecycle space where a simulation can no longer simply be assumed to remain current.

This includes: simulation retirement; simulation supersession; simulation replacement; simulation transition; successor simulation identification; transition boundaries; residual simulation use; historical simulation use; retirement conditions; retirement state; supersession state; transition state; dependency transfer; simulation lineage; evidence consequences; unresolved limitation preservation; archive status; lifecycle closure; controlled reactivation where justified.

Operational Scope

SRTM applies where continued use of an existing simulation requires explicit reconsideration because of: material reality change; Simulation Object change;

accumulated simulation drift; model obsolescence; environment obsolescence; unsupported assumptions; unavailable dependencies; obsolete input basis; inadequate scenario coverage; technological replacement; architecture change; unresolved correlation deterioration; inability to resimulate sufficiently; excessive reconstruction requirement; successor simulation introduction; termination of the original simulation purpose; loss of support or reproducibility; end of relevant operational lifecycle.

SRTM applies to individual simulation models, simulation environments, digital twins, scenario systems, simulation platforms, integrated simulation systems, mission simulations, engineering simulations, scientific simulations, safety simulations, AI simulations, autonomous-system simulations, large simulation campaigns, and future simulation technologies. Resimulation ≠ Indefinite Continuation

SLRS must not assume that every lifecycle problem can or should be solved through another simulation run.

Repeated resimulation may become inappropriate where the underlying simulation basis itself has become obsolete.

Therefore:

Resimulation Capability ≠ Continued Simulation Legitimacy

A simulation may remain technically executable while no longer being methodologically appropriate for its original purpose.

Retirement Principle

Simulation retirement occurs when a simulation ceases to hold an active governed role for its defined purpose.

Retirement may occur because: its Simulation Object no longer exists in the represented form; the object has materially changed; the original purpose has ended; its representation has become materially obsolete; assumptions can no longer be supported; the model can no longer represent relevant behavior; the environment has become obsolete; required inputs are unavailable; scenario space has changed beyond practical adaptation; correlation with reality has materially deteriorated; a successor simulation has replaced it; the simulation cannot be reconstructed sufficiently; continued use would exceed legitimate reliance boundaries.

Retirement should be explicit.

Non-use ≠ Governed Retirement.

Retirement State

SRTM establishes a formal Retired Simulation State.

A retired simulation should be distinguishable from: active simulation; temporarily suspended simulation; simulation under review; simulation undergoing resimulation; superseded simulation; archived simulation; successor simulation.

The state should indicate: retirement date; retirement basis; affected purpose; affected Simulation Object; final applicable configuration; successor where applicable; residual permitted uses; prohibited uses; evidence consequences; archival disposition.

Transition Principle

Not every simulation lifecycle ends through simple retirement.

Many simulations transition into: updated simulation versions; successor models; replacement environments; new digital twins; new mission simulators; new system configurations; redesigned simulation architectures.

SRTM governs the transition relationship between:

Outgoing Simulation Basis and Incoming Simulation Basis.

Transition must not create an unexamined assumption that the successor inherits the legitimacy of the predecessor.

Successor Simulation Principle

A successor simulation must be identifiable as a distinct governed simulation basis where material changes justify that distinction.

The relationship may be:

Version Successor

Configuration Successor

Model Successor

Environment Successor

Representation Successor

Purpose Successor

Complete Simulation Replacement

SRTM preserves lineage between predecessor and successor without treating them as methodologically identical.

Simulation Lineage ≠ Simulation Equivalence.

Transition Boundary

A transition must identify what crosses from the predecessor into the successor.

Potentially transferable elements include: Simulation Object definitions; purpose definitions; scope structures; reality representations; assumptions; model components; environmental configurations; parameter structures; input datasets; scenario libraries; execution procedures; historical outputs;

correlation records; simulation evidence; known limitations; unresolved anomalies; lifecycle records.

Transferability must be assessed rather than assumed.

Inherited Basis Principle

A successor simulation may inherit parts of an earlier simulation basis.

But inherited elements must remain identifiable as:

Inherited and Reconfirmed

Inherited with Modification

Inherited Conditionally

Not Transferable

Replaced

This prevents simulation modernization from silently carrying obsolete assumptions, limitations, scenarios, or evidence into a new simulation generation. Retirement ≠ Deletion

SRTM establishes a critical distinction:

Simulation Retirement ≠ Simulation Deletion

A retired simulation may remain valuable for: historical comparison; research; incident reconstruction; longitudinal analysis; methodological analysis; regression comparison; training; audit support; understanding previous decisions; comparison with successor simulations.

Retirement governs active legitimacy, not necessarily physical existence.

Residual Use

A retired simulation may retain explicitly bounded residual uses.

Examples include: historical reconstruction; comparison with past system versions; educational use; research; regression analysis; archival reference; legacy incident analysis.

Residual use must not recreate active reliance implicitly.

A simulation retired from operational decision support should not regain that role merely because someone can still execute it.

Residual Use Limitation

Where residual use is permitted, SRTM should define: permitted purpose; prohibited purpose; applicable object version; applicable temporal period;

known limitations; successor status; warning that the simulation is retired; whether new execution is permitted; whether outputs may enter current evidence processes.

This creates a governed distinction between:

historically useful

and

currently authoritative.

Supersession Principle

A simulation becomes Superseded where another governed simulation basis has formally replaced its active role.

Supersession should identify:

Predecessor Simulation → Successor Simulation

together with: transition date; transferred elements; non-transferred elements; changed assumptions; changed representation; changed models; changed scenarios; changed inputs; evidence consequences; residual predecessor use.

Supersession must not erase methodological history.

Transition Completeness

A transition is not complete merely because the successor simulation can execute.

Transition completeness may require confirmation that: successor identity is established; purpose and scope are established; transferred elements are known; non-transferable elements are identified; relevant assumptions are reconsidered; model and environment changes are governed; inputs remain appropriate; scenario space is sufficient; execution basis is established; outputs are distinguishable from predecessor outputs; correlation relationships are reconsidered where necessary; simulation evidence is not improperly inherited; predecessor status is updated.

Evidence Transition Principle

Simulation-derived evidence from a predecessor simulation must not automatically become evidence of the successor simulation.

A successor may materially change: representation; assumptions; model; environment; input basis; scenario structure; execution behavior; output meaning; reality correlation.

SRTM therefore requires inherited evidence to be identified for reassessment where the successor basis materially differs.

SERBS continues to govern the evidential meaning and reliance boundary.

SRTM governs the lifecycle consequence of the transition.

Historical Evidence Preservation

Retirement does not erase the historical evidential meaning that a simulation legitimately possessed at the time it was active.

Historical evidence may remain relevant to: historical decisions; previous system configurations; prior mission states; earlier operational periods; longitudinal comparison.

However:

Historically Legitimate Evidence ≠ Currently Applicable Evidence.

Retirement Due to Reality Change

A simulation may require retirement where Operational Reality has moved beyond the representational space the simulation can reasonably maintain.

Examples may include: fundamentally changed operating environments; new physical conditions; redesigned infrastructure; new system architecture; changed human behavior; changed regulatory or operational constraints; newly discovered phenomena; previously unknown interactions.

ORA™ governs Operational Reality.

SRTM governs whether the resulting change makes continued simulation use inappropriate.

Retirement Due to Model Obsolescence

A simulation may become obsolete because its underlying model no longer represents the relevant system sufficiently.

Model obsolescence may result from: new scientific knowledge; better physical models; discovered model deficiencies; unsupported software; changed system architecture; unavailable dependencies; inability to reproduce historical execution conditions.

SRTM does not determine general model quality.

It governs the lifecycle consequence where model obsolescence makes continued simulation use inappropriate.

Retirement Due to Accumulated Drift

Individual drift events may initially remain tolerable.

Over time, however:

Small Reality Changes Object Changes Input Changes Scenario Changes Model Limitations Correlation Degradation → Accumulated Simulation Drift

Accumulated drift may eventually make incremental resimulation insufficient.

SRTM provides the governance point at which the organization may determine:

stop adapting the old simulation and establish a new simulation basis.

Replacement Threshold

Organizations may define a Replacement Threshold representing the point at which continued modification of an existing simulation becomes methodologically inappropriate relative to replacement.

The threshold may consider: extent of structural change; number of affected lifecycle layers; accumulated assumptions; representation divergence; model modification burden; dependency obsolescence; scenario reconstruction requirements; correlation deterioration; inability to preserve traceability; evidence discontinuity.

The threshold should not be based solely upon economic convenience.

It is a simulation lifecycle governance decision.

Retirement with Unresolved Issues

A simulation may be retired while unresolved anomalies, limitations, discrepancies, or evidence questions remain.

These should not disappear with retirement.

SRTM requires material unresolved issues to remain attached to the retired simulation record where relevant.

This prevents retirement from functioning as a mechanism for eliminating inconvenient simulation history.

Reactivation Principle

A retired simulation may sometimes need to be reactivated.

Examples include: historical incident reconstruction; comparison against a successor; renewed research; legacy-system analysis.

Reactivation does not automatically restore the simulation's former active status.

SRTM requires the intended reactivation purpose to be explicit.

Where the simulation is intended to return to active current use, the relevant SIMULOS® lifecycle layers must be reconsidered.

Technical Reactivation ≠ Governance Reinstatement.

Simulation Lineage

SRTM requires meaningful predecessor-successor relationships to remain traceable.

A lineage may appear as:

Simulation V1 → Simulation V2 → Simulation V3 → Replacement Simulation R1 → Replacement Simulation R2

Each transition should preserve enough information to understand: what changed; why it changed; what was inherited; what was replaced; what evidence remained applicable; what evidence required reassessment; when active authority transferred.

This creates a governed simulation history rather than disconnected simulation versions.

Lifecycle Closure

A simulation lifecycle reaches governed closure when: active status has ended; retirement or supersession basis is documented; successor relationship is identified where applicable; residual use is defined; prohibited use is defined; evidence consequences are identified; unresolved material issues are preserved; archival disposition is established; lifecycle status is traceable.

Lifecycle closure does not mean that all simulation information disappears.

It means its governance state is explicit.

SRTM Boundary with SRSM

SRSM determines:

what must be resimulated.

SRTM determines:

whether continued resimulation remains appropriate, whether transition is required, or whether the simulation should retire.

Therefore:

Resimulation Scope ≠ Lifecycle Continuation Decision.

SRTM Boundary with SOS

SRTM may identify that a successor Simulation Object version or configuration requires a new simulation basis.

SOS governs the identity, boundary, version, configuration, and documentation of the Simulation Object.

SRTM governs the lifecycle transition between simulation generations.

SRTM Boundary with SMES

SRTM may determine that model or environment obsolescence requires replacement.

SMES governs the successor model and simulation environment when the lifecycle re-enters that layer.

SRTM Boundary with SES

SRTM may terminate active execution authority for a retired simulation.

SES governs simulation execution itself.

SRTM does not govern execution procedures.

SRTM Boundary with SERBS

SRTM identifies evidence potentially affected by retirement, supersession, replacement, or transition.

SERBS governs the qualification, weight, reliance boundary, use limitations, and validation interface of simulation-derived evidence.

Therefore:

Retirement does not automatically invalidate historical evidence, and transition does not automatically transfer evidential legitimacy.

SRTM Boundary with ORA™

ORA™ governs Operational Reality and changes occurring within it.

SRTM governs the simulation lifecycle consequences arising when those changes make continued simulation use inappropriate.

SRTM Boundary with VALIDOS®

SRTM does not determine whether an operational system remains validated.

A simulation's retirement may affect a validation evidence basis, but VALIDOS® determines the resulting validation consequences.

SRTM Boundary with INTEGROS®

SRTM does not govern general integrity.

INTEGROS® may govern integrity properties of archived simulation objects, records, evidence, states, and transition information.

SRTM governs the simulation-specific lifecycle status and transition.

SRTM Boundary with ADIT®

ADIT® may audit or verify retirement records, transition evidence, predecessor-successor relationships, and historical execution records.

SRTM governs the retirement and transition process itself.

SRTM Boundary with AGA™

SRTM does not determine organizational accountability for approving retirement, replacement, or continued use.

AGA™ governs authority, responsibility, accountability, and attribution.

SRTM defines the simulation-governance conditions that the lifecycle decision must address.

Minimum Implementation Framework

A minimum SRTM implementation should establish: 1. Lifecycle State Review

Determine whether the simulation remains active, requires transition, is superseded, or should retire. 2. Retirement / Transition Basis

Document the simulation-specific conditions supporting the lifecycle change. 3. Continuation Feasibility

Determine whether resimulation or modification remains methodologically appropriate. 4. Successor Identification

Identify a successor simulation where replacement or transition occurs. 5. Transferability Assessment

Determine which simulation elements may legitimately transfer to the successor and which require renewed governance. 6. Evidence Consequence Assessment

Identify simulation-derived evidence affected by retirement or transition. 7. Residual Use Definition

Define any permitted historical, analytical, research, training, or reconstruction uses. 8. Prohibited Use Definition

Identify uses that must cease after retirement or supersession. 9. Lineage & Transition Record

Preserve predecessor-successor relationships, transferred elements, material changes, and transition state. 10. Lifecycle Closure Record

Assign and document the final governed lifecycle state. Use Case 1 — Space Mission Simulation: Transition to a New Spacecraft Configuration

A space organization maintains a high-fidelity mission simulator for a spacecraft architecture used across several mission cycles.

Over time, the next spacecraft generation introduces: redesigned propulsion; new navigation hardware; different thermal behavior; modified flight software; new landing systems; changed mission phases.

The existing simulator can technically be modified again.

But accumulated changes now affect the Simulation Object, reality representation, assumptions, model structure, environment, inputs, scenarios, and previous simulation-to-reality correlations.

SRTM asks whether continuing to modify the legacy simulator remains methodologically legitimate.

The organization determines that the old simulator should become Superseded and a successor simulation basis should be established.

SRTM governs:

Legacy Simulation → Transferability Assessment → Reusable Scenario Assets → Reconfirmed Assumptions → Rejected Legacy Assumptions → New Model Components → New Correlation Requirements → Successor Simulation → Legacy Simulation Retirement.

Historical mission outputs remain available for previous mission analysis.

But they are not automatically treated as evidence for the new spacecraft generation.

The critical principle is:

A successor mission simulator may inherit knowledge from its predecessor without inheriting its legitimacy automatically. Use Case 2 — Autonomous Robot: Legacy Simulator After Major Platform Redesign

An autonomous mobile robot was originally developed and tested using a simulation environment representing its first-generation sensors, actuators, perception stack, physical dimensions, and control architecture.

Several years later, the robot platform is redesigned.

The new generation introduces: different cameras; new depth sensors; changed wheel geometry; new actuator behavior; different latency; redesigned perception software; new autonomous planning behavior.

The organization still possesses millions of historical simulation scenarios and results.

Simply continuing to use the old simulator would be computationally convenient.

But the represented robot no longer corresponds sufficiently to the new operational platform.

SRTM establishes a transition from:

Legacy Robot Simulation → Successor Robot Simulation.

Scenario libraries may remain partially transferable.

Historical failure scenarios may remain highly valuable.

Some environmental representations may remain reusable.

But sensor models, dynamics, assumptions, configurations, outputs, correlation relationships, and simulation evidence require renewed governance.

The legacy simulator is retained for historical comparison and regression research but is prohibited from serving as the current primary simulation basis for the redesigned robot.

This prevents a common lifecycle failure:

the operational system evolves, while the simulation quietly remains anchored to yesterday's system.

Governance Outputs

SRTM may produce: Simulation Lifecycle State Record Simulation Retirement Assessment Simulation Retirement Decision Record Simulation Supersession Record Simulation Transition Plan Successor Simulation Identification Record Simulation Transferability Assessment Inherited Simulation Basis Record Non-Transferable Element Record Simulation Residual Use Statement Simulation Prohibited Use Record Simulation Evidence Transition Record Historical Evidence Applicability Record Simulation Lineage Record Unresolved Issue Preservation Record Simulation Archive Disposition Simulation Reactivation Record Simulation Lifecycle Closure Record

Enterprise Application

SRTM enables organizations to govern a simulation problem that becomes increasingly important as simulation systems persist for years or decades:

When should an existing simulation stop being treated as the current simulation basis?

Without lifecycle retirement governance, organizations may continue using models, environments, assumptions, scenarios, and evidence long after the systems or realities they represent have materially changed.

SRTM enables organizations to: distinguish active, superseded, transitioned, archived, and retired simulations; determine when incremental resimulation is no longer sufficient; replace obsolete simulation foundations deliberately; preserve useful historical simulation assets; prevent obsolete simulations from supporting current decisions; establish predecessor-successor lineage; determine what can transfer into a successor; prevent obsolete assumptions from silently migrating into new simulations; preserve unresolved failures and limitations; control historical and residual use; identify evidence affected by transition; close simulation lifecycles explicitly.

This is especially important for long-lived engineering programs, mission systems, digital twins, autonomous systems, infrastructure, scientific models, and simulation platforms that evolve across multiple technological generations.

SRTM ensures that simulation governance does not end when execution stops.

It ends only when the simulation's lifecycle status, successor relationship, residual use, evidential consequences, and historical position are explicitly governed.

Foundational Principles

Resimulation does not imply indefinite continuation.

A simulation may remain executable after it ceases to be methodologically current.

Non-use is not governed retirement.

Retirement is not deletion.

Simulation lineage is not simulation equivalence.

A successor simulation does not automatically inherit the legitimacy of its predecessor.

Transferable simulation assets must be distinguished from transferable simulation authority.

Historically legitimate evidence is not automatically currently applicable evidence.

Accumulated drift may require replacement rather than another incremental update.

Technical reactivation does not automatically restore governance status.

Obsolete simulations must not remain operationally authoritative merely because they remain accessible.

The end of a simulation lifecycle must be governed as explicitly as its beginning.

Canonical Closing Statement

Simulation Retirement & Transition Module establishes the final lifecycle-control layer of the Simulation Lifecycle & Resimulation Standard by governing when simulations cease active use, become superseded, transition into successor simulation bases, retain bounded residual uses, or reach formal lifecycle closure. Through governance of retirement conditions, continuation feasibility, supersession, successor identification, transferability, inherited simulation bases, residual and prohibited uses, evidence consequences, historical applicability, unresolved issues, reactivation, archival disposition, and simulation lineage, SRTM prevents obsolete or superseded simulations from retaining methodological authority merely because they remain technically available. By distinguishing retirement from deletion, lineage from equivalence, historical evidence from current applicability, and technical reactivation from governance reinstatement, SRTM closes the recursive SIMULOS® lifecycle while preserving the knowledge necessary for future simulation generations. A governed simulation lifecycle is complete only when not merely its creation and execution, but also its transition, supersession, residual use, and retirement have become explicit, traceable, and methodologically bounded.

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